



Technical data sheet

XT-ATTO-LXL



XT-ATTO-LXL

XT-ATTO-LXL is a very small embedded network module, which can be easily integrated into any end device. The XT-ATTO-LXL embedded network module has the dimensions of 26 x 30 x 3mm, can be installed via a 67-pin M.2 PCI connector and can therefore be used directly or as an option in your circuit. A total of up to 10 interfaces are supported simultaneously such as UART(RS232, RS485), I2C, SPI, QSPI, CAN, Ethernet, PWM, ADC, MDIO, USB, LCD, ISC, PDMIC as well as TTL-IO. With only a few additional components, such as an RJ45 jack with integrated transformer and a 5 Volt power supply, your terminal device becomes immediately network-compatible and accessible worldwide. As OS Linux with kernel version 5.4.81-linux4sam is preinstalled.

Hardware – Description



Emulations and functions

- Connect-On-Data
- Auto-Connect
- SSH-Zugriff
- FTP-Server
- TCP/UDP -Client
- TCP/UDP -Server
- TTL – IO
- USB 2.0
- UART
- QSPI
- 128 MB RAM
- 4/8/16 GB eMMC-Speicher
- 100Mbit Ethernet
- SD-Card
- I2C – Master
- SPI – Master / Slave
- CAN
- ADC
- ARM Cortex-A5 500MHz
- I2S, ISC

Supported systems

1. Windows
2. Linux
3. UNIX

Management

1. SSH /Telnet
2. Browser
3. Serial interface

Technical data

Temperature range:
-40°C.. + 85°C

Standards:
CE / WEEE / RoHS
EN 55022 Class B
EN 55024 Class A

Power supply:
5 Volt
200 mA

Dimensions:
26 x 30 x 3 mm

Weight:
3 Gramm

Ethernet (MDIX)
10 Half Duplex
10 Full Duplex
100 Half Duplex
100 Full Duplex
AutoSensing

Interfaces - Properties

- All data pins 3.3Volt TTL, 10K pullup
- All data interfaces freely selectable

5 x UART(RS232/RS485)
4 x I2C
4 x SPI
1 x MDIO
1 x USB-Host 2.0/USB-Device
1 x Ethernet max. 100 Mbit/s
1 x CAN
2 x I2S
1 x QSPI
1 x PDMIC
1 x ISC
1 x LCD
51 x GPIO

Supported protocols IP Dual Stack

- | | |
|----------------|------------------|
| 1. IPv4 | 20. IPv6 |
| 2. TCP | 21. NDP |
| 3. UDP | 22. ICMPv6 |
| 4. FTP | 23. DHCPv6 |
| 5. TFTP | 24. TCPv6 |
| 6. ICMP | 25. UDPv6 |
| 7. ARP | 26. Netbios-NS |
| 8. SNMP | 27. LLMNR |
| 9. LPR | 28. ZeroConfig |
| 10. DHCP | - APIPA |
| 11. BOOTP | - AutoIP |
| 12. DNS | 29. IP-Multicast |
| 13. TELNET | 30. AK-M2M |
| 14. HTML | 31. SSL/TLS |
| 15. HTTP/HTTPS | |
| 16. DYNDNS | |
| 17. SMTP | |
| 18. POP3 | |
| 19. SYSLOG | |

Functions

Operating system

The module is run with Linux as operating system. You get ssh access for an administrator user and can use it to develop/install/configure your own programs. The image includes opkg package manager. So you can install almost any Linux package which is executable on an ARM architecture. If you wish, you can also get root access for our image or build your own Linux image.



Functions

Virtual-USB

The AK-Nord Linux image (without root access) includes a license for Virtual USB solution. This allows almost all USB devices connected to the module (USB host) to be shared on the network. You only need the free Virtual-USB client installed on your PC and you can run a USB device, which is accessible anywhere in the network, on the PC.

For the PC it looks as if the USB device is directly connected to the PC.

Shell

You get SSH access to the module and can use your own shell scripts.

Webserver

You can develop your own administration web interface. Either you use the pre-installed webserver lighttpd or you develop your own webserver.

Python3

The AK-Nord image also includes the Python3 interpreter. Thus you can use your own python programs.

C/C++ Applications

With an ARM Cross Compiler you can also develop your own software modules in C/C++.

Updatesystem

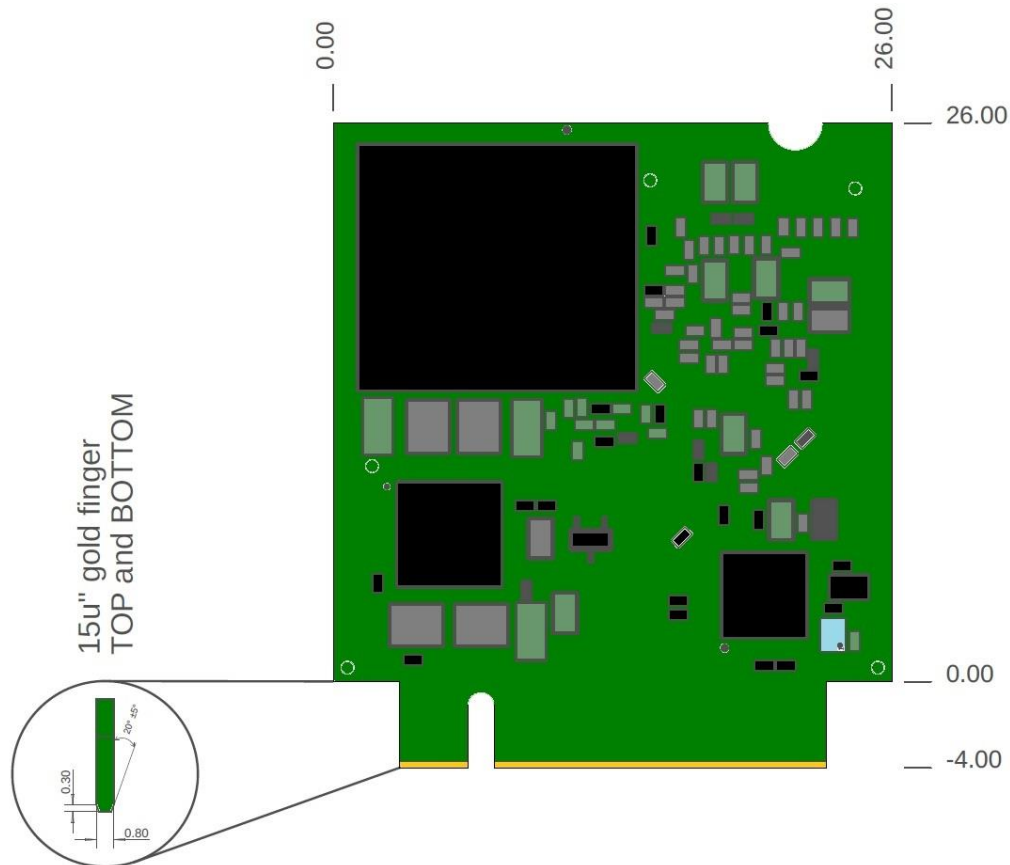
The AK-Nord image is equipped with an update system and can be updated completely or in parts via browser.

Rescue-System

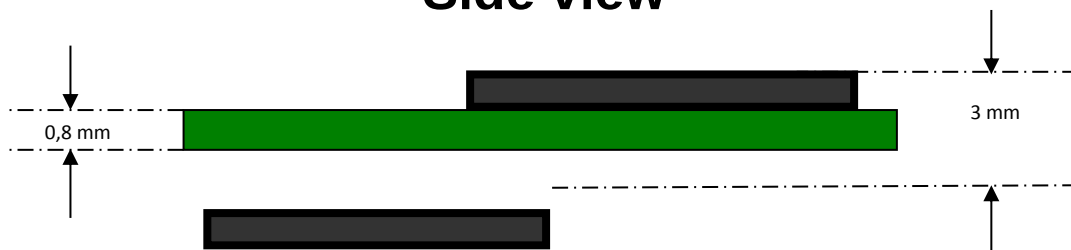
If a module is no longer bootable (misadministration/power failure during an update), a rescue image can be started via a button/jumper, which can be accessed via a fixed IP in network. This can be used to reset a module to factory settings.

Dimensions

Top view

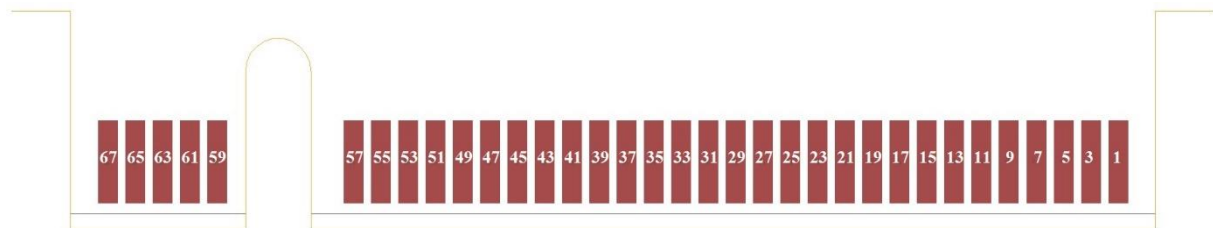


Side view

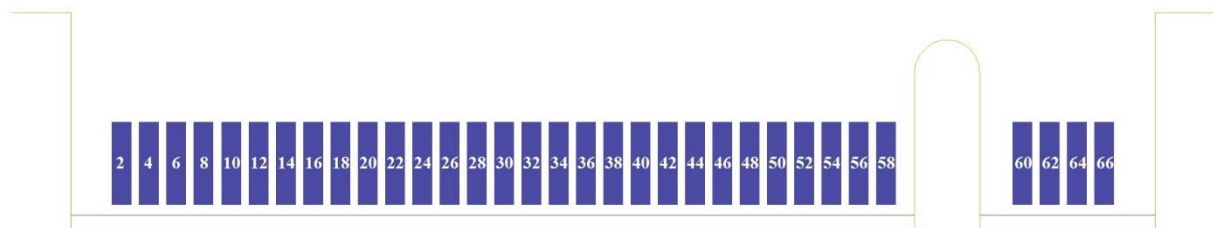


PIN - Description

Top view



Bottom view



Absolute Maximum Ratings

Ambient temperature under bias	-40°C to +85°C
Storage temperature	-65°C to +150°C
Voltage on VDD	-0.3V to +5.0V
Voltage on VDDBU	-0.3V to +3.3V
Voltage on any 3.3 V pin	-0.3V to 3.6V

PIN - Description

Non-configurable pins:

M.2 PIN	GPIO PIN	Description
60	VDD	Power supply 5V0
66	GND	Power supply GND
64	VDDBU	Power supply backup battery 3V3
43	NRST	Reset
11	LED_LINK	Link LED Phyter
1	PC26	ACT LED/Status LED
9	ETH_RX_N	Ethernet Phyter RX-N
7	ETH_RX_P	Ethernet Phyter RX-P
5	ETH_TX_N	Ethernet Phyter TX-N
3	ETH_TX_P	Ethernet Phyter TX-P
62	PC28	Rescuemodus Button/Factoryreset
67		Wakeup
51	USBBP	USB PortB P
49	USBBM	USB PortB M
47	PD2	Debug UART RXD
45	PD3	Debug UART TXD

Configurable interfaces:

Interface 1:

M.2 PIN	GPIO PIN	Version 1_1	QSPI1 1_2	PWM 1_3
4	PB5	V4	QSPI1_SCK	PWMH2
2	PB6	V3	QSPI1_CS	PWML2
6	PB7	V0	QSPI1_IO0	PWMH3
8	PB8	V1	QSPI1_IO1	PWML3
10	PB9	V2	QSPIO_IO2	PWMFI1
12	PB10	V5	QSPIO_IO3	PWMEXTRG2

PIN - Description

Interface 2 (2_0 - 2_8):

M.2 PIN	GPIO PIN	LCD 2_0	ISC 2_1	SPI1 2_3	PDMIC 2_4	UART3 2_5	QSPI1 2_6	I2SC1 2_7	FC3 USART 2_8u	FC3 SPI 2_8s	FC3 I2C 2_8i
44	PB11	DAT0			DAT	URXD3					
36	PB12	DAT1			CLK	UTXD3					
40	PB13	DAT2									
32	PB14	DAT3					SCK	MCK			
30	PB15	DAT4					CS	CK			
28	PB16	DAT5					IO0	WS			
26	PB17	DAT6					IO1	DI0			
24	PB18	DAT7					IO2	DO0			
22	PB19	DAT8					IO3				
20	PB20	DAT9									
18	PB21	DAT10							SCK	SPCK	
16	PB22	DAT11							RXD	MISO	TWCK
14	PB23	DAT12							TXD	MOSI	TWD
34	PB24	DAT13	D10						CTS	NPCS0	
48	PB25	DAT14	D11						RTS	NPCS1	
25	PB26	DAT15	D0								
23	PB27	DAT16	D1								
21	PB28	DAT17	D2								
19	PB29	DAT18	D3								
17	PB30	DAT19	D4								
15	PB31	DAT20	D5								
13	PC0	DAT21	D6								
58	PC1	DAT22	D7	SCK							
56	PC2	DAT23	D8	MOSI							
54	PC3	PWM	D9	MISO							
52	PC4	DISP	PCK	NPCS0							
50	PC5	VSYN	VSYN	NPCS1							
42	PC6	HSYN	HSYN	NPCS2							
38	PC7	PCK	MCK	NPCS3							
46	PC8	DEN	FIELD								

PIN - Description

Interface 2 (2_9 – 2_14):

M.2 PIN	GPIO PIN	PDMIC 2_9	UART0 2_10	FC0 USART 2_11u	FC0 SPI 2_11s	FC0 I2C 2_11i	I2C0 2_12	CAN0 2_13	I2SC0 2_14
44	PB11								
36	PB12								
40	PB13								
32	PB14								
30	PB15								
28	PB16								
26	PB17								
24	PB18								
22	PB19								
20	PB20								
18	PB21								
16	PB22								
14	PB23								
34	PB24								
48	PB25								
25	PB26	DAT	URXD0						
23	PB27	CLK	UTXD0						
21	PB28			TXD	MOSI	TWD			
19	PB29			RXD	MISO	TWCK			
17	PB30			SCK	SPCK				
15	PB31			CTS	NPCS0		TWD0		
13	PC0			RTS	NPCS1		TWCK0		
58	PC1							CANTX0	I2SC0_CK
56	PC2							CANRX0	I2SC0_MCK
54	PC3								I2SC0_WS
52	PC4								I2SC0_DI0
50	PC5								I2SC0_DO0
42	PC6								
38	PC7								
46	PC8								

Interface 3:

M.2 PIN	GPIO PIN	SPI1 3_1	FC2- SUART 3_2u	FC2- SPI 3_2s	FC2- I2C 3_2i	UART2 RX/TX 3_3	UART2 7PIN 3_3f	I2C0 3_4	ADC 3_5
41	PD23					URXD2	RXD		AD4
39	PD24					UTXD2	TXD		AD5
37	PD25	SPI1_SPCK					RTS		AD6
35	PD26	SPI1_MOSI	TX	MOSI	TWD		CTS		AD7
33	PD27	SPI1_MISO	RX	MISO	TWCK		DTR		AD8
31	PD28	SPI1_NPCS0	SCK	SPCK			DSR		AD9
29	PD29	SPI1_NPCS1	CTS	NPCS0			DCD	TWD0	AD10
27	PD30	SPI1_NPCS2	RTS	NPCS1				TWCK0	AD11

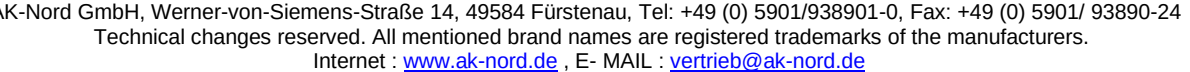


PIN - Description

Interface 4:

M.2 PIN	GPIO PIN	SDMMC1 4_1
59	PA18	SDMMC1_DAT0
57	PA19	SDMMC1_DAT1
55	PA20	SDMMC1_DAT2
53	PA21	SDMMC1_DAT3
65	PA22	SDMMC1_CK
63	PA28	SDMMC1_CMD
61	PA30	SDMMC1_CD

Schematic



Schematic

